

# Preventive Maintenance Procedures — Rotating Equipment

## 1. Purpose

This document defines the preventive maintenance (PM) schedule and procedures for rotating equipment including pumps (P-1xx series) and compressors (C-2xx series) in the process area.

## 2. Pump Maintenance Schedule

The following maintenance intervals apply to all P-100 series pumps unless otherwise specified in the equipment-specific manual:

- Daily: visual inspection, check for leaks, unusual noise, or vibration.
- Weekly: record bearing temperature and casing temperature readings.
- Monthly: check oil level and oil condition; verify cooling water flow.
- Quarterly: vibration analysis with portable analyzer, compare to baseline.
- Annually: full bearing inspection, oil change, shaft alignment check.

## 3. Temperature Trending

Bearing and casing temperatures should be logged at every round and trended over time. A gradual increase in bearing temperature over several days, even if still below the alarm threshold, is an early indicator of developing bearing wear or lubrication degradation and should trigger an inspection before the alarm is reached.

## 4. Work Order Procedure for High Temperature Alarms

When a high temperature alarm is received from any rotating equipment:

- Operator confirms the alarm at the control room and logs the reading.
- Field operator performs a walk-down inspection within 30 minutes.
- If temperature exceeds the trip setpoint, equipment is shut down per the safety manual.
- Maintenance is notified and a work order is raised for root-cause investigation.
- Equipment-specific manual (e.g. pump manual, compressor manual) is consulted for likely causes before opening the equipment.

## 5. Spare Parts

Critical spares for P-100 series pumps (bearings, mechanical seals, coupling elements) should be maintained in the warehouse at a minimum stock level of 2 units per part number.